

6. Setting up your breadboard

G. 5 amp

$$V=IR$$

$$I = V/R = 5V/1 \text{ Ohm}$$

$$I = 5 \text{ amp}$$

H. No this is not sufficient because the resistor is too weak.

I. The board would probably break.

K. The higher the resistance, the lower the current will be. This prevents overheating and breaking the breadboard.

L. Fluctuating between 2 - 5 V, this is what I expected. The multimeter measured around 0.010 to 0.015 amps, which I also expected.

7. Building a real RC circuit

F. $T = 1 \text{ sec}++$

$$T = RC = 10,000 * 0.0001 = 1 \text{ sec}$$

I. Yes, at first the voltage rises quickly. However, after a few seconds it starts to slow down significantly

K. It takes about 1 or 2 seconds for the voltage to drop ~37% of where it started. This is around 1 to 2 time constants ($T = 1$ from part F)

L. using $R = 1 \text{ k Ohm}$, $R = 22 \text{ k Ohm}$ (same as group for comparison)

For $R = 1 \text{ k Ohm}$, $C = 220 \text{ uF}$

$$T = RC = 0.22 \text{ sec}$$

For $R = 22 \text{ k Ohm}$, $C = 220 \text{ uF}$

$$T = RC = 4.85 \text{ sec} \rightarrow \sim 5 \text{ seconds}$$

M. For the 1 k Ohm resistor, it was a little hard to tell how long it took to get to 5V, I would say around a second or two. For the 22 k Ohm resistor, it was in line with the experimental data.